Chapter 2: Asset Classes and Financial Instruments

Multiple Choice

1. A Treasury bill with a face value of $10,000 and 120 days to maturity is selling for $9,780. What are its bank discount yield (BDY) and bond equivalent yield (BEY)?
a. BDY 6.60%; BEY 6.84%
b. BDY 6.84%; BEY 6.60%
c. BDY 6.69%; BEY 6.84%
d. BDY 6.60%; BEY 6.69%

2. For the same Treasury bill, which ordering of the three yield measures is correct?
a. Effective annual yield < bond equivalent yield < bank discount yield
b. Bond equivalent yield < bank discount yield < effective annual yield
c. Bank discount yield < bond equivalent yield < effective annual yield
d. All three are equal because they describe the same bill

3. Which of the following money market instruments is quoted on an interest-bearing (add-on) basis rather than on a discount basis?
a. Negotiable certificate of deposit
b. Treasury bill
c. Commercial paper
d. Banker's acceptance

4. In a repurchase agreement (repo), the dealer:
a. Buys securities from an investor with an agreement to sell them back later at a lower price
b. Sells securities to an investor with an agreement to buy them back the next day at a higher price
c. Lends securities to an investor in exchange for an unsecured promissory note
d. Exchanges dollar deposits for foreign-currency deposits at an agreed forward rate

5. A municipal bond yields 3.2% and a taxable corporate bond of comparable risk yields 4.5%. For an investor in the 32% marginal tax bracket, what is the equivalent taxable yield of the municipal bond, and which bond is better?
a. Equivalent taxable yield 4.71%; the corporate bond
b. Equivalent taxable yield 2.18%; the corporate bond
c. Equivalent taxable yield 4.71%; the municipal bond
d. Equivalent taxable yield 4.22%; the municipal bond

6. A taxable corporate bond yields 7.0% and a comparable-risk municipal bond yields 4.2%. At what marginal tax rate would an investor be indifferent between the two?

a. 28%

b. 35%

c. 40%

d. 60%

7. A Treasury bond with a par value of $1,000 is quoted at an asked price of 103-12. What is the dollar price of the bond?

a. $1,031.20

b. $1,031.88

c. $1,033.75

d. $1,037.50

8. A TIPS bond has a par value of $1,000 and a coupon rate of 2.0%. Inflation over the year is 3.5%, and the principal is adjusted accordingly. What coupon payment does the holder receive for the year?

a. $20.00

b. $20.70

c. $35.00

d. $55.70

9. Which pair of characteristics best describes common stock?

a. Fixed claim and unlimited liability

b. Residual claim and unlimited liability

c. Residual claim and limited liability

d. Fixed claim and limited liability

10. Preferred stock is often called a hybrid of debt and equity because:

a. It can be converted into the firm’s bonds at the holder’s discretion at any time and pays interest instead of dividends

b. It carries full voting rights like common stock and a guaranteed maturity date like a bond

c. It pays a fixed dividend like a perpetual bond with no maturity, yet it is equity that ranks below bondholders and above common stockholders

d. Its dividend is tax-deductible for the issuer, like interest, and rises automatically with the firm’s earnings

11. A foreign company's common share trades at KRW 540,000 in Seoul. One ADR represents 1/10 of a common share, and the exchange rate is KRW 1,350 per U.S. dollar. Ignoring costs, what is the ADR's approximate dollar price?

a. $4.00

b. $40.00

c. $400.00

d. $54,000

12. Consider the three stocks in the table, where Pt is the price and Qt the shares outstanding at time t. Stock C splits two-for-one in the last period (its price falls from 132 to 70 after the split). What is the rate of return on a price-weighted index for the first period (t = 0 to t = 1), and what must the divisor become for the second period so that the split does not change the index level?

Stock	P0	Q0	P1	Q1	P2	Q2
A	80	150	86	150	89	150
B	45	300	42	300	40	300
C	120	100	132	100	70	200

a. Return 3.20%; new divisor 2.24

b. Return 6.12%; new divisor 3.00

c. Return 6.12%; new divisor 2.24

d. Return 6.12%; new divisor 2.38

13. Using the three stocks in the table (stock C splits two-for-one in the last period), what are the first-period returns (t = 0 to t = 1) on an equally weighted index and on a value-weighted index?

Stock	P0	Q0	P1	Q1	P2	Q2
A	80	150	86	150	89	150
B	45	300	42	300	40	300
C	120	100	132	100	70	200

a. Equally weighted 3.61%; value-weighted 3.20%

b. Equally weighted 3.20%; value-weighted 3.61%

c. Equally weighted 3.61%; value-weighted 6.12%

d. Equally weighted 6.12%; value-weighted 3.20%

14. Which statement about an equally weighted index is correct?
a. It corresponds to a buy-and-hold strategy, so no trading is needed as prices change
b. It gives the greatest weight to the highest-priced stocks, as the Dow Jones Industrial Average does
c. It gives the greatest weight to the largest-capitalization firms, as the S&P 500 does
d. Maintaining it requires periodic rebalancing — selling stocks that have risen and buying stocks that have fallen

15. Which statement about a call option is correct?
a. The holder is obligated to buy the asset at expiration regardless of the asset's price
b. The holder has the right but not the obligation to buy the asset at the exercise price, and loses at most the premium paid
c. The holder's loss is unlimited, because the asset price can rise without bound and the holder must deliver at the strike
d. A call is exercised only when the market price falls below the strike price at or before expiration

16. You buy a call option with a strike price of $50 for a premium of $3 per share, and the stock trades at $57 at expiration. You also buy a put option with a strike price of $60 for a premium of $4. What is your profit per share on the call, and at what stock price does the put break even?
a. Call profit $7; put break-even $56
b. Call profit $4; put break-even $64
c. Call profit $10; put break-even $60
d. Call profit $4; put break-even $56

17. Which statement correctly distinguishes a long futures position from a long call position?
a. The futures buyer has the right but not the obligation to take delivery, while the call buyer is obligated to buy
b. Both positions require an up-front premium payment of the same size
c. The futures buyer's maximum loss is limited to the initial margin posted
d. The futures buyer is obligated to settle at the futures price, while the call buyer will exercise only if it is profitable

Chapter 2: Asset Classes and Financial Instruments — Answers

1. a
BDY = [(F − P) ÷ F] × (360 ÷ n) = (220 ÷ 10,000) × (360 ÷ 120) = 6.60%; BEY = [(F − P) ÷ P] × (365 ÷ n) = (220 ÷ 9,780) × (365 ÷ 120) = 6.84%.

2. c
BDY divides by face value over 360 days, BEY divides by the lower purchase price over 365 days, and EAY also compounds, so accuracy and size rise BDY → BEY → EAY.

3. a
T-bills, commercial paper and bankers' acceptances are sold below face value (discount basis), whereas CDs, repos, federal funds and Eurodollar deposits pay principal plus interest at maturity (add-on).

4. b
A repo is a short-term, often overnight, collateralized loan to the dealer; the higher repurchase price is the interest, and in a reverse repo the investor lends cash and holds the collateral.

5. c
Equivalent taxable yield = 3.2% ÷ (1 − 0.32) = 4.71%, which exceeds the corporate bond's 4.5%, so the municipal bond leaves more after tax (3.2% versus 4.5% × 0.68 = 3.06%).

6. c
Set after-tax yields equal: 7.0% × (1 − t) = 4.2%, so t = 1 − 4.2 ÷ 7.0 = 40%; investors above this bracket prefer the muni, those below prefer the taxable bond.

7. c
The digits after the dash are 32nds of a point: 103 + 12/32 = 103.375% of par, i.e. $1,033.75 per $1,000.

8. b
TIPS pay the coupon on the inflation-adjusted principal: $1,000 × 1.035 = $1,035, and $1,035 × 2.0% = $20.70.

9. c
Shareholders are last in line, receiving only what remains after creditors and preferred holders are paid, and they can lose no more than the amount invested because personal assets are shielded.

10. c
Preferred pays a fixed, often cumulative dividend and has priority over common but ranks below debt; unlike bond interest, its dividends are not tax-deductible for the issuer.

11. b
ADR price = share price × ratio ÷ exchange rate = 540,000 × 1/10 ÷ 1,350 = $40.00.

12. c
Price-weighted return = (86 + 42 + 132) ÷ (80 + 45 + 120) − 1 = 260 ÷ 245 − 1 = 6.12%; just before the split the index is (86 + 42 + 132) ÷ 3 = 86.67, and after it the sum is 86 + 42 + 66 = 194, so the divisor must be d = 194 ÷ 86.67 = 2.24.

13. a
Equally weighted = average of the three returns = (7.50% − 6.67% + 10.00%) ÷ 3 = 3.61%; value-weighted = total market value growth = (86×150 + 42×300 + 132×100) ÷ (80×150 + 45×300 + 120×100) − 1 = 38,700 ÷ 37,500 − 1 = 3.20%.

14. d
An equally weighted index averages stock returns, i.e. an equal dollar amount in each stock; prices drift the weights apart, so the investor must keep trading back to equal dollars.

15. b
The buyer pays a premium for a right: the loss is capped at the premium and the gain is open-ended, and the call is exercised only when the market price is above the strike.

16. d
Call profit = max(57 − 50, 0) − 3 = $4; a put pays K − S_T, so the premium is recovered when S_T = K − premium = 60 − 4 = $56.

17. d
An option is a right, bought for a premium, with a capped loss; a futures contract is an obligation with no premium (margin only), marked to market daily, so a move against you must still be settled.

Chapter 3: Securities Markets

Multiple Choice

1. Which of the following transactions takes place in the primary market?
a. An investor buys 100 shares of Samsung Electronics from another investor on the exchange
b. A mutual fund sells its holdings of Toyota stock to a pension fund
c. A firm sells newly issued shares to the public in an initial public offering
d. A dealer buys shares into inventory from a customer

2. In a firm commitment underwriting, the investment bank:
a. Acts only as an agent and returns any unsold shares to the issuer
b. Guarantees the issuing firm a minimum share price for one year after the offering
c. Buys the entire issue from the firm and bears the placement risk of reselling it to the public
d. Lends the issuing firm the offering proceeds, to be repaid out of future earnings

3. A firm sells 200,000 shares in an initial public offering at an offer price of $25. The underwriter’s explicit fees are $150,000, and the share price closes at $28 on the first day of trading. What is the best estimate of the total cost of the equity issue to the firm?
a. $150,000
b. $600,000
c. $750,000
d. $4,850,000

4. The sell side of a limit-order book for a stock shows the orders in the table. You submit a market order to buy 700 shares. What total amount will you pay, ignoring commissions?

Ask price	Shares offered
$31.40	300
$31.45	500
$31.55	600

a. $21,980
b. $22,000
c. $22,015
d. $22,085

5. Which statement correctly contrasts a market order with a limit order?
a. A market order guarantees a price but not execution; a limit order guarantees execution but not price
b. A market order executes immediately at the best available price; a limit order executes only at the specified price or better, and may never execute
c. Both orders execute immediately at the quoted price, but a limit order pays a lower commission
d. A limit order can only be used to buy, while a market order can be used to buy or sell and is the only order allowed during trading hours

6. You own a stock at $120 and place a stop-loss order at $108. Overnight, bad news makes the stock open the next morning at $102. What happens?
a. The shares are sold at exactly $108, because a stop-loss order guarantees the stop price
b. The order is never triggered, because the price skipped over $108 without trading at it
c. The stop is triggered and a market order sells the shares near the $102 opening price
d. The order becomes a limit order at $108, which cannot be filled at the lower opening price

7. Which statement correctly contrasts the NYSE and Nasdaq?
a. The NYSE is an auction, order-driven market with a central limit-order book, while Nasdaq is a dealer, quote-driven market with competing market makers
b. The NYSE is a dealer market with competing market makers, while Nasdaq is an auction market with a central limit-order book
c. Both are pure dealer markets with no limit-order book
d. Nasdaq has a physical trading floor, while the NYSE has no human oversight at all

8. You have $15,000 of cash and the initial margin requirement is 40%. What is the maximum value of stock you can buy, and how much do you borrow? (Ignore commissions.)
a. Buy $37,500; borrow $15,000
b. Buy $37,500; borrow $22,500
c. Buy $30,000; borrow $15,000
d. Buy $25,000; borrow $10,000

9. An investor buys 800 shares at $25 per share using the maximum allowable initial margin of 50%, so the margin loan is $10,000. The maintenance margin is 35%. At what share price will the investor receive a margin call?
a. $8.75
b. $12.50
c. $16.25
d. $19.23

10. An investor buys 300 shares at $60 per share, borrowing 40% of the purchase price at an interest rate of 7% per year. One year later the shares are sold at $66. The stock pays no dividends and commissions are ignored. What is the rate of return on the investor’s equity?

a. 9.67%

b. 10.00%

c. 12.00%

d. 16.67%

11. Which statement about buying stock on margin is correct?

a. Margin reduces the volatility of the investor’s return because the loan is a fixed obligation

b. Margin magnifies both gains and losses on the investor’s equity, and interest on the loan accrues whatever the stock does

c. The investor’s maximum loss on a margined position equals the loss on an unmargined position

d. The initial margin requirement is set by each individual broker and cannot be changed by regulators

12. Which statement about the cash flows and obligations in a short sale is correct?

a. The proceeds are held in the account as collateral, additional margin must be posted, and the short seller must reimburse any dividends the stock pays

b. The short seller receives the sale proceeds in cash and may withdraw them immediately to invest elsewhere, with no further obligation

c. The short seller is entitled to receive any dividends paid on the borrowed shares, because the seller is the owner of record

d. The position requires no margin because the shares were sold rather than bought, so no capital is at risk

13. An investor sells short 200 shares at $40 per share and deposits the required initial margin of 50% in cash. The maintenance margin is 35%. At what share price will the investor receive a margin call?

a. $39.00

b. $44.44

c. $54.00

d. $60.00

14. An investor sells short 500 shares at $60 per share and deposits an initial margin of 50% of the short value. One year later the investor covers at $54, after the stock paid a dividend of $1.50 per share. Treating the initial margin as the amount invested, what is the rate of return on the short position?

a. 7.50%

b. 15.00%

c. 20.00%

d. 25.00%

15. Which statement about U.S. securities regulation is correct?

a. The Securities Exchange Act of 1934 tightened governance after the 2002 scandals, and the CFTC regulates futures

b. The Sarbanes–Oxley Act of 2002 tightened auditing and governance standards after accounting scandals, and the CFTC regulates futures

c. The Sarbanes–Oxley Act of 2002 tightened governance, and FINRA is the government agency that regulates futures

d. The Securities Act of 1933 tightened governance after the 2002 scandals, and the SEC regulates futures

Chapter 3: Securities Markets — Answers

1. c
The primary market is where new securities are issued and the issuing firm receives the proceeds; every later trade among investors is a secondary-market trade that brings the firm no new money.

2. c
The underwriter sets an offer price from book-building demand and, under a firm commitment, bears the placement risk; its compensation is the spread between the price it pays and the price it charges.

3. c
Cost = explicit fee + money left on the table = \$150,000 + (28 − 25) × 200,000 = \$750,000.

4. b
A market buy fills against the lowest asks first: 300 × \$31.40 + 400 × \$31.45 = \$22,000 (an average of \$31.43).

5. b
A market order buys certainty of execution but not of price; a limit order buys certainty of price but not of execution, and you rarely get both.

6. c
A stop-loss guarantees execution, not price, so after a gap down it fills near the opening price; a stop-limit order, by contrast, would guarantee the price but might not fill at all.

7. a
The NYSE uses a central limit-order book with price–time priority (with human oversight at the open and close); Nasdaq has no trading floor and liquidity comes from multiple dealers posting quotes.

8. b
Buying power = cash ÷ initial margin = \$15,000 ÷ 0.40 = \$37,500; the loan is \$37,500 − \$15,000 = \$22,500.

9. d
Margin = (800P − 10,000) ÷ 800P = 0.35 → 800P × 0.65 = 10,000, so P = \$19.23; below this price a margin call is issued.

10. c
Position = 300 × \$60 = \$18,000; equity = \$10,800, loan = \$7,200; ending equity = 300 × \$66 − \$7,200 × 1.07 = \$12,096, so the return is 12,096 ÷ 10,800 − 1 = 12.00% (versus 10.00% unlevered).

11. b
The shares are collateral for a fixed loan, so all price variation falls on the investor's equity; leverage scales the return up and down, and Reg T sets initial margin at a minimum of 50% equity.

12. a
Short sales must be done in a margin account: the proceeds are not withdrawn, the investor also deposits initial margin, and dividends and borrow fees are costs of staying short.

13. b
Collateral = proceeds \$8,000 + margin \$4,000 = \$12,000; margin = (12,000 − 200P) ÷ 200P = 0.35, so P = 12,000 ÷ (200 × 1.35) = \$44.44; for a short position the call comes when the price rises.

14. b
Profit = (60 − 54 − 1.50) × 500 = \$2,250 (the dividend must be reimbursed); the investment is the margin of \$15,000, so the return is 2,250 ÷ 15,000 = 15.00%.

15. b
Sarbanes–Oxley (2002) responded to the accounting scandals; the SEC oversees securities and security options, the CFTC oversees futures, and FINRA is the industry self-regulatory body.

Chapter 4: Mutual Funds and Other Investment Companies

Multiple Choice

1. Which of the following is NOT one of the four core services an investment company provides to small investors?
a. Diversification and divisibility
b. Lower transaction costs from trading in large blocks
c. Record keeping and administration
d. A guaranteed return above the market index

2. A fund holds 120,000 shares of A at $52, 200,000 shares of B at $18, 90,000 shares of C at $64, and 350,000 shares of D at $9.50. Liabilities are $0.9 million and there are 2.2 million fund shares outstanding. What is the NAV per share?
a. $6.68
b. $8.19
c. $8.60
d. $9.01

3. An open-end fund has a NAV of $32.00. An investor buys $1.6 million of the fund. How many new shares are issued, and what is NAV afterward?
a. 50,000 shares; NAV rises to $32.80
b. 1,600,000 shares; NAV stays at $32.00
c. No new shares; NAV rises
d. 50,000 shares; NAV stays at $32.00

4. You place an order to buy shares of an open-end mutual fund at 10:30 a.m. At what price is the order filled?
a. At the last NAV published before 10:30 a.m.
b. At the market price quoted on the exchange at 10:30 a.m.
c. At any limit price you specify, if the market reaches it
d. At the NAV struck at the 4:00 p.m. close that day

5. Which statement about closed-end funds is correct?
a. Shares outstanding are fixed after the IPO, and the market price can differ from NAV
b. Investors redeem shares directly with the fund at NAV whenever they wish
c. Shares outstanding change daily as investors buy new shares from, and redeem them to, the fund
d. The market price is required by law to equal NAV at the close of each trading day

6. You buy a closed-end fund at $21.00 when its NAV is $24.00. One year later NAV is still $24.00, the price is $22.50, and the fund distributed $1.05 per share. What is your rate of return?
a. 4.38%
b. 7.14%
c. 11.33%
d. 12.14%

7. Which feature describes a unit investment trust (UIT)?
a. An actively managed portfolio overseen by a board of directors
b. A private pool open only to accredited investors
c. A fixed, unmanaged portfolio that terminates on a set date
d. A fund that trades continuously on an exchange at NAV

8. Compared with a mutual fund, a hedge fund typically:
a. Offers daily redemption at NAV to the general public, like an open-end mutual fund
b. Faces tighter limits on leverage and short sales because it is regulated more heavily
c. Is open only to accredited or institutional investors and is largely exempt from the Investment Company Act
d. Charges a lower fee than a mutual fund because it is less regulated and pays no incentive fee

9. Which statement about 12b-1 fees is correct?
a. They are one-time charges paid when shares are purchased, and are deducted from the amount invested
b. They are annual distribution fees of up to 1% of assets and work like a sales load spread over time
c. They are paid to portfolio managers as a performance bonus when the fund beats its benchmark
d. They are charged only on redemption within the first five years, and decline by one point annually

10. A fund reports a one-year return of 9%. For an investor who bought Class A shares with a 4% front-end load, the reported return:
a. Understates her return, because loads are refunded after one year
b. Equals her return, because loads are included in the expense ratio
c. Equals her return, because loads are deducted from NAV
d. Overstates her return, because published returns are computed before loads

11. You invest $30,000 in a fund with a 6% front-end load. How much is actually invested in the fund, and what return must the fund earn for you to get back to $30,000?
a. $28,200 invested; 6.00% return needed
b. $28,302 invested; 6.00% return needed
c. $31,800 invested; 5.66% return needed
d. $28,200 invested; 6.38% return needed

12. You invest $20,000 in Class A shares with a 4% front-end load. The fund’s reported return for the year is 12.0%. What is your return?

a. 7.52%

b. 8.00%

c. 11.54%

d. 12.00%

13. A fund charges a CDSC of 5%, 4%, 3%, 2% and 1% in years 1–5 and 0% thereafter. You redeem shares worth $22,000 during year 4. If the charge applies to the redemption value, what is it?

a. $220

b. $440

c. $660

d. $880

14. A fund’s NAV rises from $24.60 to $25.90 during the year. It pays income distributions of $0.45 and capital gain distributions of $0.95 per share, and its expense ratio is 1.25%. What are its net return and approximate gross return?

a. Net 10.98%; gross 12.23%

b. Net 5.28%; gross 6.53%

c. Net 10.42%; gross 11.67%

d. Net 10.98%; gross 9.73%

15. You invest $10,000 in a fund with a gross return of 8% a year. The two share classes are shown in the table. Which class leaves you with more after 1 year, and which after 10 years?

Feature	Class A	Class C
Front-end load	4.5%	none
Expense ratio	0.75%	1.45%
CDSC	none	1% if redeemed in year 1

a. 1 year: Class A ($10,242); 10 years: Class A ($19,230)

b. 1 year: Class C ($10,548); 10 years: Class C ($18,860)

c. 1 year: Class C ($10,548); 10 years: Class A ($19,230)

d. 1 year: Class A ($10,725); 10 years: Class C ($18,860)

16. Why do ETF holders generally avoid the capital gains distributions that mutual fund holders receive when other investors redeem?
a. ETFs are exempt from capital gains tax by law, so no gains are ever distributed
b. ETFs hold only bonds, which by their nature do not produce capital gains
c. ETF sellers trade with other investors on the exchange, so the fund does not have to sell stock to raise cash
d. ETF managers never sell securities for any reason, not even to rebalance the index

17. An ETF trades at $75.60 while its underlying basket is worth $75.00 per ETF share. Creation units are 20,000 shares. Ignoring trading costs, what is the AP's profit from one creation, and what does the AP do?
a. $12,000 — buy ETF shares, redeem them, sell the basket
b. $12,000 — buy the basket, create ETF shares, sell them
c. $1,200 — buy the basket, create ETF shares, sell them
d. $24,000 — buy ETF shares, redeem them, sell the basket

18. An index moves +8%, −8%, +8%, −8% on four consecutive days, starting at 100. A 2× daily leveraged ETF also starts at 100. What are the ending values?
a. Index 98.72; 2× ETF 94.95
b. Index 98.72; 2× ETF 97.45
c. Index 100.00; 2× ETF 100.00
d. Index 98.72; 2× ETF 98.72

19. You own a stock at $40 and sell a call with a $44 strike for a $1.20 premium. What is your profit per share on the covered call if the stock ends at $50, and if it ends at $35?
a. $10.00 at $50; −$5.00 at $35
b. $11.20 at $50; $1.20 at $35
c. $5.20 at $50; −$3.80 at $35
d. $5.20 at $50; −$6.20 at $35

20. Why must the average actively managed dollar underperform the market after costs?
a. Active managers are, on average, less skilled than the managers who build index funds
b. Before costs, active investors as a group hold the market, so their average gross return equals the market return
c. Active funds hold more cash than the market does, which drags returns in rising years
d. Regulations force active funds to hold riskier stocks than the index contains

21. Of 164 U.S. large-cap funds in the top quartile in 2021, 20.1% were still in the top quartile in 2022 and 0.0% in 2023, while pure chance would give 6.25%. What does this suggest?

a. Ranking funds by past returns is a good way to choose a fund

b. Past top performance is not a reliable signal of future top performance

c. Top funds deliberately take less risk after a good year

d. The result is caused by survivorship bias inflating the winners

Chapter 4: Mutual Funds and Other Investment Companies — Answers

1. d
The four services are record keeping and administration, diversification and divisibility, professional management and lower transaction costs; whether managers succeed on average is an empirical question.

2. b
Assets = 120,000×52 + 200,000×18 + 90,000×64 + 350,000×9.5 = $18,925,000; NAV = (18,925,000 − 900,000) ÷ 2,200,000 = $8.19.

3. d
New shares = $1,600,000 ÷ $32.00 = 50,000; assets and shares rise in proportion, so NAV stays at $32.00.

4. d
Open-end funds transact only at NAV, struck once daily at 4:00 p.m. ET, so every order that day gets the same price and you do not know it when you order; intraday prices and limit orders belong to ETFs and closed-end funds.

5. a
A closed-end fund neither issues nor redeems shares after its IPO; investors trade with each other on an exchange, so supply and demand set a price that may be at a premium or discount to NAV.

6. d
Return = (22.50 − 21.00 + 1.05) ÷ 21.00 = 12.14%, measured on the market price actually paid.

7. c
A sponsor deposits a fixed portfolio into a trust and sells redeemable units; with essentially no trading afterwards, management fees are very low and the trust is self-liquidating.

8. c
Hedge funds are private pools that disclose little, may use short sales, derivatives and heavy leverage, impose lock-up periods, and charge about “2 and 20”.

9. b
Named for SEC Rule 12b-1, this annual charge (maximum 1% of assets) pays for marketing and adviser compensation; it is included in the expense ratio but economically it is a load paid in instalments.

10. d
Reported returns are computed from NAV, which is net of expenses but not of loads, so a load-fund investor earns less than the published figure.

11. d
The load is a percentage of the gross amount paid: $30,000 × (1 − 0.06) = $28,200; to recover $30,000 the fund must earn 30,000 ÷ 28,200 − 1 = 6.38%, because the load is 6% of the gross but 6.38% of the net amount invested.

12. a
Amount invested = 20,000 × 0.96 = $19,200; end value = 19,200 × 1.12 = $21,504; return = 21,504 ÷ 20,000 − 1 = 7.52%.

13. b
The year-4 charge is 2%: 2% × $22,000 = $440.

14. a
Net return = (25.90 − 24.60 + 0.45 + 0.95) ÷ 24.60 = 10.98%, since distributions must be added back and the denominator is the beginning NAV; gross return ≈ net return + expense ratio = 10.98% + 1.25% = 12.23%.

15. c
Net growth: A = 8% − 0.75% = 7.25%, C = 8% − 1.45% = 6.55%; after 1 year A = 10,000 × 0.955 × 1.0725 = $10,242 and C = 10,000 × 0.99 × 1.0655 = $10,548 (the 1% CDSC applies in year 1); after 10 years A = 10,000 × 0.955 × 1.0725^{10} = $19,230 and C = 10,000 × 1.0655^{10} = $18,860.

16. c
In a mutual fund a redemption forces the fund to sell stock and realize gains that land on holders who sold nothing, whereas an ETF seller simply finds a buyer on the exchange.

17. b
The ETF is rich relative to the basket, so the AP buys the basket and creates ETF shares to sell; profit = (75.60 − 75.00) × 20,000 = $12,000.

18. a
Index: 100 × 1.08 × 0.92 × 1.08 × 0.92 = 98.72 (−1.28%); 2× ETF: 100 × 1.16 × 0.84 × 1.16 × 0.84 = 94.95 (−5.05%), about four times the index’s loss rather than twice.

19. c
At $50 the stock is called away at $44: profit = (44 − 40) + 1.20 = $5.20, so the upside above $44 was sold for the premium; at $35 the call expires worthless: profit = (35 − 40) + 1.20 = −$3.80, because the premium only softens the loss by $1.20 and is not protection.

20. b
Passive investors hold the market in proportion, so what remains for active investors is also the market; subtract their higher fees and the average must lag, though this says nothing about any single fund.

21. b
If skill drove rankings, winners would repeat more often than chance; the evidence shows no persistence, so past winners are not a dependable guide.

Chapter 5: Risk, Return, and the Historical Record

Multiple Choice

1. You buy a share at KRW 52,000. One year later it trades at KRW 57,200, and you received a cash dividend of KRW 1,300. What is the holding-period return?
a. 2.50%
b. 10.00%
c. 11.36%
d. 12.50%

2. A fund returns +30%, −25%, +15% and −10% in four consecutive years. What are its arithmetic and geometric average returns?
a. Arithmetic 2.50%, geometric 2.50%
b. Arithmetic 0.23%, geometric 2.50%
c. Arithmetic 2.50%, geometric 0.91%
d. Arithmetic 2.50%, geometric 0.23%

3. At t = 0 you buy one share at KRW 50,000. At t = 1 the share pays a KRW 1,500 dividend and you buy a second share at KRW 56,000. At t = 2 each share pays a KRW 1,500 dividend and you sell both shares at KRW 54,000. What are your dollar-weighted and time-weighted (geometric) returns?

Time	Share price	Dividend per share	Trade
t = 0	KRW 50,000	–	Buy 1 share
t = 1	KRW 56,000	KRW 1,500	Buy 1 more share
t = 2	KRW 54,000	KRW 1,500	Sell both shares

a. Dollar-weighted 4.15%; time-weighted 6.76%
b. Dollar-weighted 6.76%; time-weighted 4.15%
c. Dollar-weighted 4.15%; time-weighted 7.05%
d. Dollar-weighted 15.00%; time-weighted 6.76%

4. A deposit quotes 9% compounded monthly. What is the effective annual rate?
a. 0.75%
b. 9.00%
c. 9.31%
d. 9.38%

5. An investment has an effective annual rate of 12%. What is the equivalent continuously compounded annual rate?

a. 11.33%

b. 11.39%

c. 12.00%

d. 12.75%

6. A bond earns a nominal return of 7.5% while inflation is 3.2%. What is the exact real return?

a. 4.08%

b. 4.17%

c. 4.30%

d. 10.94%

7. A portfolio returns 40% in a boom (probability 0.25), 12% in a normal year (0.45), and −18% in a bust (0.30). What are its expected return and standard deviation?

State	Probability	Return
Boom	0.25	40%
Normal	0.45	12%
Bust	0.30	−18%

a. E(r) = 10.00%, σ = 21.49%

b. E(r) = 10.00%, σ = 4.62%

c. E(r) = 11.33%, σ = 23.68%

d. E(r) = 10.00%, σ = 41.09%

8. Two return distributions have the same mean and standard deviation. Distribution N is symmetric, while distribution L has negative skew. Which statement about L is correct?

a. It has a long right tail, so upside surprises are more frequent than in N

b. Its mean, median and mode coincide, as they do in the symmetric distribution N

c. It has a long left tail (mean < median < mode), so the standard deviation understates its downside risk

d. Its standard deviation overstates the risk of large losses, since the left tail is long

9. Compared with a normal distribution having the same mean and standard deviation, a return distribution with positive excess kurtosis has:

a. Thinner tails, so extreme returns are rarer than normality predicts

b. A long left tail only, so the mean lies below the median and mode

c. A taller peak and heavier tails on both sides, so extreme returns occur more often than normality predicts

d. A larger standard deviation by construction, since kurtosis is defined from the variance

10. A portfolio worth ₩5 billion has a daily volatility of 1.4%. Assuming normal returns, what are its one-day 95% VaR and its 10-day 99% VaR? (Use z = 1.645 for 95% and z = 2.326 for 99%.)
a. ₩115.15 million and ₩162.82 million
b. ₩115.15 million and ₩364.14 million
c. ₩115.15 million and ₩1,628.20 million
d. ₩115.15 million and ₩514.88 million

11. A bank reports a one-day 95% VaR of ₩30 million. Which statement is correct?
a. The largest loss the bank can suffer in one day is ₩30 million
b. The bank expects to lose ₩30 million per day on average
c. On 95 days out of 100, the bank will lose exactly ₩30 million
d. On about 5 trading days in 100, the loss is expected to exceed ₩30 million

12. Which statement correctly distinguishes value at risk (VaR) from expected shortfall (ES)?
a. VaR measures the average loss beyond the cut-off, while ES only marks the cut-off at a given confidence level
b. VaR marks the cut-off loss at a confidence level, while ES measures how large the loss is on average once that cut-off is passed
c. VaR and ES always give the same number when returns are normally distributed at any confidence level
d. ES ignores losses beyond the cut-off, which is why regulators prefer it to VaR for capital rules

13. Fund A has E(r) = 14% and σ = 24%; Fund B has E(r) = 9% and σ = 10%. The risk-free rate is 3%. Which fund has the higher Sharpe ratio?
a. Fund B: 0.90 versus 0.58
b. Fund A: 0.46 versus 0.60, because it earns more
c. Fund A: 2.18 versus 1.67
d. Fund B: 0.60 versus 0.46

14. As the investor changes y (the fraction of wealth in risky portfolio P, with the rest in the risk-free asset), what happens to the Sharpe ratio of the complete portfolio?
a. It is unchanged for every y
b. It rises as y rises
c. It falls as y rises
d. It rises until y = 1 and then falls

15. Over 1928–2021 the S&P 500 had an arithmetic average return of 11.82% and 3-month T-bills averaged 3.38%. What is the equity risk premium over T-bills?
a. 3.50%
b. 6.60%
c. 8.44%
d. 15.20%

16. Risky portfolio P has E(rP) = 15% and σP = 24%; the risk-free rate is 5%. An investor places y = 0.8 of her wealth in P and the rest in the risk-free asset. What are the expected return, standard deviation and Sharpe ratio of the complete portfolio?

a. E(rC) = 13.00%; σC = 24.00%; Sharpe 0.33

b. E(rC) = 13.00%; σC = 19.20%; Sharpe 0.42

c. E(rC) = 12.00%; σC = 19.20%; Sharpe 0.36

d. E(rC) = 13.00%; σC = 19.20%; Sharpe 0.68

17. Wealth is KRW 30 million, y = 0.6, and risky portfolio P is 55% equities and 45% bonds. How is the money actually allocated?

a. Equities KRW 16.5m, bonds KRW 13.5m, T-bills KRW 0.0m

b. Equities KRW 9.9m, bonds KRW 8.1m, T-bills KRW 12.0m

c. Equities KRW 18.0m, bonds KRW 0.0m, T-bills KRW 12.0m

d. Equities KRW 6.6m, bonds KRW 5.4m, T-bills KRW 18.0m

18. The portfolios in the table are available to investors with utility $U = E(r) - \tfrac{1}{2} \cdot A \cdot \sigma^2$ and A = 2, 5 and 10. Which portfolio does each investor choose?

Portfolio	E(r)	σ
Conservative	6%	4%
Moderate	10%	12%
Aggressive	15%	22%

a. A = 2: Conservative; A = 5: Moderate; A = 10: Aggressive

b. All three investors choose Aggressive (highest E(r))

c. A = 2: Moderate; A = 5: Moderate; A = 10: Aggressive

d. A = 2: Aggressive; A = 5: Moderate; A = 10: Conservative

19. Portfolio Z has E(r) = 14% and σ = 24%. An investor with A = 4 can instead hold a 3% T-bill. What is Z’s certainty equivalent, and what does she hold?

a. Certainty equivalent 2.48%; she holds the T-bill

b. Certainty equivalent 2.48%; she holds Z because E(r) = 14% is higher

c. Certainty equivalent −9.04%; she holds the T-bill

d. Certainty equivalent 25.52%; she holds Z

20. E(rP) = 12%, σP = 20%, rf = 3% and the investor’s risk aversion is A = 4. What are the optimal risky weight y*, and the resulting E(rC), σC and utility $U = E(r) - \tfrac{1}{2}A\sigma^2$?

a. y* = 1.1250; E(rC) = 13.12%; σC = 22.50%; U = 3.00%

b. y* = 0.5625; E(rC) = 8.06%; σC = 11.25%; U = 5.53%

c. y* = 0.1125; E(rC) = 4.01%; σC = 2.25%; U = 3.91%

d. y* = 0.5625; E(rC) = 8.06%; σC = 11.25%; U = 8.06%

21. Risky portfolio P has E(rP) = 12% and σP = 20%; the risk-free rate is 3%. An investor with A = 2 has optimal weight y* = 1.125 and wealth of KRW 16 million. Assuming she can borrow at the risk-free rate, what does she do?

a. Invests KRW 2.0m in P and holds the rest in T-bills

b. Borrows KRW 18.0m and invests KRW 34.0m in P

c. Borrows KRW 2.0m and invests KRW 18.0m in P

d. Holds KRW 18.0m in P and KRW 2.0m in T-bills

22. Holding everything else fixed, which change would make the optimal risky weight y* smaller?

a. A rise in the risk premium E(rP) − rf

b. A fall in the coefficient of risk aversion A

c. A fall in the variance of the risky portfolio

d. A rise in the variance of the risky portfolio

23. In $U = E(r) - \frac{1}{2} \cdot A \cdot \sigma^2$, which statement about A is correct?

a. A < 0 describes a risk-averse investor

b. A = 0 describes a risk-neutral investor who judges by E(r) alone

c. A larger A means the investor is more tolerant of risk

d. A has no effect when σ is below 20%

24. Risky portfolio P has E(rP) = 15% and σP = 25%; the risk-free rate is 5%. What is the slope of the capital allocation line, and where does it cross σ = 0?

a. Slope 0.60; intercept 5%

b. Slope 0.40; intercept 5%

c. Slope 0.40; intercept 0%

d. Slope 2.50; intercept 15%

25. What is the capital market line (CML)?

a. The line joining the lowest-risk and highest-return securities in the market

b. The capital allocation line of the best-performing active fund over the sample period

c. A line showing the mean and variance of every individual stock in the market

d. The capital allocation line that uses a broad market index as the risky portfolio P

Chapter 5: Risk, Return, and the Historical Record — Answers

1. d
HPR = (57,200 − 52,000 + 1,300) ÷ 52,000 = 12.50%, i.e. capital gain yield 10.00% plus dividend yield 2.50%.

2. d
Arithmetic = (30 − 25 + 15 − 10) ÷ 4 = 2.50%; geometric = (1.30 × 0.75 × 1.15 × 0.90)^(1/4) − 1 = 0.23%, always below the arithmetic mean.

3. a
Net cash flows: t0 = −50,000; t1 = −56,000 + 1,500 = −54,500; t2 = 2 × 1,500 + 2 × 54,000 = +111,000, whose IRR is the dollar-weighted return, 4.15%. Period returns: year 1 = (56,000 − 50,000 + 1,500) ÷ 50,000 = 15.00%, year 2 = (54,000 − 56,000 + 1,500) ÷ 56,000 = −0.89%, so the time-weighted return = (1.15 × 0.99107)^(1/2) − 1 = 6.76%; the dollar-weighted return is lower because more money was invested in the weaker second year.

4. d
EAR = $(1 + 0.09/12)^{12} - 1$ = 9.38%.

5. a
APR (continuous) = ln(1 + EAR) = ln(1.12) = 11.33%.

6. b
Fisher equation: 1 + R = (1 + r)(1 + i), so r = 1.075 ÷ 1.032 − 1 = 4.17%.

7. a
E(r) = 0.25(40) + 0.45(12) + 0.30(−18) = 10.00%; σ^2 = $0.25(30)^2 + 0.45(2)^2 + 0.30(-28)^2$ = 462, so σ = √462 = 21.49%.

8. c
With negative skew the left tail is long and the mean falls below the median and mode; equity returns are typically left-skewed, so σ alone understates their risk.

9. c
Fat tails mean extremes arrive far more often than a normal distribution implies (Black Monday 1987 was about a −20σ day), so normality understates the chance of crashes.

10. d
VaR = V0 × z × σ × √T. One-day 95%: ₩5,000 million × 1.645 × 0.014 × 1 = ₩115.15 million; 10-day 99%: ₩5,000 million × 2.326 × 0.014 × √10 = ₩514.88 million.

11. d
VaR is the loss exceeded with probability 5% over the stated horizon; it is not a maximum loss and says nothing about how large the loss is on those worst days.

12. b
VaR says nothing about the size of losses past the cut-off; ES (also CVaR) answers that and is coherent, and Basel FRTB replaced the 99% VaR charge with a 97.5% expected shortfall.

13. d
Sharpe = (E(r) − rf) ÷ σ: A = (14 − 3) ÷ 24 = 0.46; B = (9 − 3) ÷ 10 = 0.60, so B gives more reward per unit of volatility.

14. a
Both the risk premium and σC scale with y, so their ratio stays equal to the slope of the CAL; this is why choosing P and choosing y can be solved separately (the separation property).

15. c
The equity risk premium is 11.82% − 3.38% = 8.44 percentage points.

16. b
E(rC) = rf + y[E(rP) − rf] = 5% + 0.8 × 10% = 13.00%; σC = y·σP = 0.8 × 24% = 19.20%; Sharpe = (13.00 − 5) ÷ 19.20 = 0.42, identical to P's 10 ÷ 24, because the risk premium and σC both scale with y.

17. b
T-bills = 30m × 0.4 = KRW 12.0m; P = 30m × 0.6 = KRW 18.0m, of which equities = 18.0 × 0.55 = KRW 9.9m and bonds = 18.0 × 0.45 = KRW 8.1m.

18. d
Utilities (A = 2 / 5 / 10): Conservative 5.84 / 5.60 / 5.20; Moderate 8.56 / 6.40 / 2.80; Aggressive 10.16 / 2.90 / −9.20, so more risk aversion means a safer choice.

19. a
U = 0.14 − ½ × 4 × 0.24^2 = 2.48%, below the 3% certain return of the T-bill, so she holds the T-bill; a risk-free asset has σ = 0, so its certainty equivalent is its own return.

20. b
y* = [E(rP) − rf] ÷ (A·σP^2) = 0.09 ÷ (4 × 0.04) = 0.5625; E(rC) = 3% + 0.5625 × 9% = 8.06%; σC = 0.5625 × 20% = 11.25%; U = 0.0806 − ½ × 4 × 0.1125^2 = 5.53%.

21. c
y* = 1.125 > 1: she invests 1.125 × 16m = KRW 18.0m in P, financed by her own 16.0m plus KRW 2.0m borrowed at the risk-free rate.

22. d
y* = [E(rP) − rf] ÷ (A·σP^2): a bigger reward invites a bigger bet, while more risk aversion or more variance shrinks it.

23. b
A > 0 is risk averse (empirically about 2 to 4), A = 0 is risk neutral and A < 0 is risk loving; the penalty ½·A·σ^2 grows with both A and σ.

24. b
Slope = (E(rP) − rf) ÷ σP = (15 − 5) ÷ 25 = 0.40, the Sharpe ratio, and the line starts at rf = 5% where σ = 0.

25. d
A passive strategy holds a broad market index without security analysis or market timing; when that index is P, the resulting CAL is called the CML.

Chapter 6: Optimal Risky Portfolios

Multiple Choice

1. Each of n stocks has a standard deviation of 30% and every pair has a correlation of 0.25. An investor holds them with equal weights. What is the portfolio standard deviation when n = 20, and what does it converge to as n grows without limit?
a. 15.00% at n = 20; floor 16.09%
b. 6.71% at n = 20; floor 0.00%
c. 16.09% at n = 20; floor 7.50%
d. 16.09% at n = 20; floor 15.00%

2. Which of the following is an example of firm-specific (diversifiable) risk?
a. The central bank unexpectedly raises interest rates for the whole economy
b. A pandemic lowers the earnings of nearly all listed firms at once
c. A pharmaceutical company announces that its main drug failed its clinical trial
d. The won depreciates sharply against the dollar, hurting importers and exporters alike

3. A stock has a total return variance of 0.0400 (σ = 20%), of which 0.0256 (σ = 16%) is systematic. What is the standard deviation attributable to firm-specific risk?
a. 1.44%
b. 4.00%
c. 12.00%
d. 25.61%

4. Four equally likely states produce the returns in the table. What are the covariance (in %²) and correlation between the stock and bond returns?

State	Prob.	Stock return	Bond return
Severe recession	0.25	−20%	+11%
Mild recession	0.25	+4%	+5%
Normal growth	0.25	+16%	+2%
Boom	0.25	+30%	+2%

a. Cov = −64.5; ρ = −0.96
b. Cov = 64.5; ρ = 0.96
c. Cov = −27.0; ρ = −0.40
d. Cov = −64.5; ρ = −2.93

5. A stock fund has E(r) = 11% and σ = 22%; a bond fund has E(r) = 5% and σ = 8%; their correlation is 0.3. What are the expected return and standard deviation of a portfolio with 70% in stocks and 30% in bonds?

a. E(rP) = 9.20%, σP = 17.80%

b. E(rP) = 9.20%, σP = 15.59%

c. E(rP) = 8.00%, σP = 12.78%

d. E(rP) = 9.20%, σP = 16.28%

6. A stock fund (E(r) = 14%, σ = 30%) and a bond fund (E(r) = 6%, σ = 10%) have a correlation of −1. What weight in stocks creates a zero-risk portfolio, and what is its expected return?

a. 75.0% in stocks; E(r) = 12.0%

b. 25.0% in stocks; E(r) = 8.0%

c. 33.3% in stocks; E(r) = 8.7%

d. 50.0% in stocks; E(r) = 10.0%

7. Stock fund σS = 24%, bond fund σB = 8%, correlation 0.1. What is the weight in stocks of the minimum-variance portfolio, and its standard deviation?

a. 10.00% in stocks; σ = 7.81%

b. 92.55% in stocks; σ = 22.28%

c. 7.45% in stocks; σ = 8.00%

d. 7.45% in stocks; σ = 7.79%

8. A portfolio is 100% in a bond fund with σB = 12%. A stock fund has σS = 30%. For which correlation with the bond fund does adding a small amount of stock reduce the portfolio's risk?

a. ρ = 0.25

b. ρ = 0.40

c. ρ = 0.60

d. ρ = 0.90

9. As the correlation between two risky assets decreases while the expected returns and standard deviations stay the same, what happens to the portfolio opportunity set?

a. It becomes a straight line between the two assets

b. It shifts upward, so every mix earns a higher expected return

c. It bows further to the left, so a given expected return can be earned at lower risk

d. It is unchanged, because expected returns are linear in the weights

10. On the opportunity set of two risky assets, which portfolios are efficient?

a. Those on the upward-sloping part starting at the minimum-variance portfolio

b. Those on the downward-sloping part below the minimum-variance portfolio

c. Every portfolio on the curve

d. Only the two single-asset portfolios

11. Stock fund: E(r) = 13%, σ = 24%; bond fund: E(r) = 5%, σ = 9%; correlation 0.25; rf = 2%. What are the weights, expected return, standard deviation and Sharpe ratio of the optimal risky portfolio?

a. Stocks 39.13%, bonds 60.87%; E(rP) = 8.13%, σP = 12.00%, Sharpe 0.511

b. Stocks 4.92%, bonds 95.08%; E(rP) = 5.39%, σP = 8.93%, Sharpe 0.380

c. Stocks 34.02%, bonds 65.98%; E(rP) = 7.72%, σP = 11.23%, Sharpe 0.509

d. Stocks 60.87%, bonds 39.13%; E(rP) = 9.87%, σP = 15.86%, Sharpe 0.496

12. rf = 2%. Which of the candidate risky portfolios in the table gives the steepest capital allocation line?

Candidate portfolio	w in stocks	E(rP)	σP
All bonds	0.00%	5.00%	9.00%
Minimum-variance portfolio	4.92%	5.39%	8.93%
60 / 40 portfolio	60.00%	9.80%	15.69%
All stocks	100.00%	13.00%	24.00%
Optimal risky portfolio	39.13%	8.13%	12.00%

a. Minimum-variance portfolio (lowest risk)

b. All stocks (highest expected return)

c. All bonds (simplest low-risk choice)

d. Optimal risky portfolio (highest Sharpe ratio)

13. The optimal risky portfolio has E(rP) = 8.13%, σP = 12.00% with 39.13% in the stock fund; rf = 2%. What fraction of total wealth should an investor with A = 5 hold in the stock fund?

a. 14.81%

b. 33.33%

c. 39.13%

d. 85.19%

14. Two clients with risk aversion A = 3 and A = 8 face the same inputs and can lend and borrow at the same risk-free rate. According to the separation property:

a. They hold the same stock-to-bond ratio inside the risky portfolio but different amounts of T-bills

b. They hold different risky portfolios, with the more risk-averse client holding more bonds inside P

c. They hold the same total allocation to risky assets

d. The client with A = 8 holds no risky assets

15. Which of the following is a reason why the separation property breaks down in practice?

a. All investors face identical taxes and the same constraints

b. Investors agree on the input list and hold the same expectations

c. Human capital is the same for every investor in the economy

d. Borrowing rates exceed lending rates, which kinks the CAL

16. Which statement about the Markowitz efficient frontier of n risky assets is correct?
a. Each individual asset lies on the efficient frontier, since single securities are its building blocks
b. Individual assets lie inside the frontier, and only the branch above the minimum-variance portfolio is efficient
c. The entire frontier is efficient, including the part below the minimum-variance portfolio
d. The tangency portfolio is found by minimizing variance for a zero expected return

17. To run the Markowitz optimization on n = 200 securities, how many estimates are needed in total, and how many of them are covariances?
a. 19,900 in total; 19,900 covariances
b. 20,300 in total; 39,800 covariances
c. 20,300 in total; 19,900 covariances
d. 40,000 in total; 19,900 covariances

18. Korean households hold mostly domestic equities, although Korea is a small share of world market capitalization. Why would adding foreign equities usually reduce risk?
a. Foreign equities always earn higher expected returns than domestic equities
b. Correlations between countries are exactly zero, so foreign risk is fully independent
c. Adding foreign equities removes market risk entirely, because currencies offset equity moves
d. Foreign markets have different industrial structures, so their correlation with domestic stocks is below +1

Chapter 6: Optimal Risky Portfolios — Answers

1. d
Average covariance = $0.25 \times 30^2 = 225$; variance = $(1/20) \times 900 + (1 - 1/20) \times 225 = 258.75$, so $\sigma = 16.09\%$; as $n \to \infty$ the first term vanishes and $\sigma \to \sqrt{225} = 15.00\%$, the systematic risk that diversification cannot remove.

2. c
Firm-specific risk arises from events affecting one company or industry and cancels out in a large portfolio; interest rates, pandemics and exchange rates hit every firm at once, so they are market (systematic) risk.

3. c
Var(total) = Var(systematic) + Var(firm-specific) because the two parts are uncorrelated, so firm-specific variance = $0.0400 - 0.0256 = 0.0144$ and $\sigma = \sqrt{0.0144} = 12.00\%$.

4. a
E(S) = 8% and E(B) = 5%; Cov = Σ p × (dev S) × (dev B) = −64.5; $\sigma S = 18.35\%$ and $\sigma B = 3.67\%$, so $\rho = -64.5 \div (18.35 \times 3.67) = -0.96$; the stock does worst exactly when the bond does best.

5. d
$E(rP) = 0.7(11) + 0.3(5) = 9.20\%$; variance = $0.49(484) + 0.09(64) + 2(0.7)(0.3)(52.8) = 265.10$, so $\sigma P = 16.28\%$.

6. b
Set $wS\sigma S = wB\sigma B$: $30w = 10(1 - w)$, so $w = 10 \div (30 + 10) = 0.25$; $E(r) = 0.25(14) + 0.75(6) = 8.0\%$, with zero standard deviation.

7. d
$Cov = 0.1 \times 24 \times 8 = 19.2$; $wS = (\sigma B^2 - Cov) \div (\sigma S^2 + \sigma B^2 - 2Cov) = (64 - 19.2) \div (576 + 64 - 38.4) = 0.0745$; $\sigma = 7.79\%$, below the bond fund's own 8%.

8. a
Adding the riskier asset lowers risk whenever $\rho < \sigma B / \sigma S = 12/30 = 0.40$; $\rho = 0.25$ satisfies this, while $\rho = 0.40$ is exactly the break-even and higher correlations raise risk.

9. c
Expected return depends only on the weights, so diversification shows up as a leftward bow; at $\rho = +1$ the set is a straight line, and at $\rho = -1$ it touches the vertical axis.

10. a
Points below the minimum-variance portfolio are dominated by points above it with the same risk but a higher expected return, so only the upper branch is efficient.

11. a
Cov = 54.0; RS = 11, RB = 3; $wS^* = [11(81) - 3(54.0)] \div [11(81) + 3(576) - 14(54.0)] = 729 \div 1863 = 0.3913$; $E(rP) = 0.3913(13) + 0.6087(5) = 8.13\%$; $\sigma P^2 = 143.93$, so $\sigma P = 12.00\%$; Sharpe = $(8.13 - 2) \div 12.00 = 0.511$.

12. d
Sharpe ratios: All bonds 0.333; Minimum-variance portfolio 0.380; 60 / 40 portfolio 0.497; All stocks 0.458; Optimal risky portfolio 0.511; the CAL slope is the Sharpe ratio, and neither the lowest-risk nor the highest-return portfolio maximizes it.

13. b
$y^* = [E(rP) - rf] \div (A \cdot \sigma P^2) = 0.0613 \div (5 \times 0.014393) = 0.8519$; stocks = $y^* \times wS^* = 0.8519 \times 0.3913 = 33.33\%$ of total wealth (look-through).

14. a
The tangency portfolio is the same for everyone; only y^* differs, so the more risk-averse client simply puts more in T-bills while keeping the identical mix within P.

15. d
The other limits listed in the slides are tax differences, human capital (a semiconductor engineer should tilt away from semiconductors), disagreement over the input list, and binding constraints.

16. b
The frontier is found by minimizing variance for each target return; single assets are dominated, and the tangency portfolio P* remains the unique Sharpe-maximizing portfolio.

17. c
Total = n expected returns + n variances + $n(n - 1)/2$ covariances = 200 + 200 + 19,900 = 20,300 = $n(n + 3)/2$; covariances are symmetric, so only $n(n - 1)/2 = 19{,}900$ are distinct.

18. d
The benefit is a covariance argument, not a forecasting argument: as long as cross-border correlations are below +1 the interaction term helps, although correlations have risen over recent decades and currency risk must be handled separately.